ARMAAN GUHA

Noida, India | +91 8377897338 | armaanguha@gmail.com
armaanguha.com | github.com/ogshrug | linkedin.com/in/armaan-guha-750223267

Why Computer Science + Cyber Security

I am drawn to cybersecurity because of the questions I already chase in my own projects - how do you tell a real signal from an anomaly, how do you protect a system before it fails, how do you make critical infrastructure trustworthy, are at their core, security questions? I have been tinkering with computers ever since I was in the 5th grade and I think I have finally found my true passion in Cybersecurity.

EDUCATION

Delhi Public School, Noida - Class 12, Science Stream (Physics, Chemistry, Math, Computer Science, English, AI)
SAT - 1500 **IELTS** - Overall 8.0 (Listening 9.0, Reading 7.5, Writing 6.5, Speaking 9.0)

PROJECTS

AcousticLeakNet - Simulation-trained deep learning system for acoustic water pipe leak detection

- Trained on 100.5M simulated data rows across 8 water distribution networks; reached perfect AUROC/F1 on 3 unseen test networks
- Low-cost two-sensor hardware design, 10–42x cheaper than commercial leak-detection alternatives
- Co-authored technical research paper; GENIUS Olympiad 2026 (AI category) Finalist - one of 27 projects from India
- Cleared Stage 2 of the Stockholm Junior Water Prize (India) and Phase 1 of the Conrad Challenge

Ferrite - Lightweight AI model runtime built from scratch in C, with zero dependencies

- Runs trained neural networks in a 222 KB program, 83x smaller than ONNX Runtime, with matching results; about 2.3x slower
- Columbia Junior Science Journal - one of 250 entries selected from 1,200 in the preliminary round
- Reproduced 97.52% accuracy on MNIST; runs AcousticLeakNet's model in under 1 MB of memory; built for Raspberry Pi Zero W and ESP32

YEMU - Malware analysis sandbox for Linux and Windows (GPCSSI)

- Runs samples in disposable QEMU VMs (KVM on Linux, WHPX on Windows) with strace behavioural monitoring, network capture, and static + in-memory YARA scanning
- Explainable 0–100 threat score and verdict; Qt desktop app and CLI, shipped as v0.6.1 with installers and CI

Log Anomaly Detector - AI pipeline for detecting abnormal system log behaviour (GPCSSI)

- Collaborative project combining an LSTM sequence model, explainable AI (SHAP), and automated log parsing (Drain3) to flag anomalous event sequences
- Built a web frontend for uploading logs and visualizing detected anomalies

KrishiMitra - Digital assistant for Indian farmers - 3rd among 50 DPS branches, Inter DPS Environment Fest

- Multilingual dashboard for soil health analysis, AI crop recommendations, weather forecasts and market prices, fed by low-cost ESP32 soil moisture, nutrient and pH sensors

armaanguha.com - Personal portfolio and blog website

- Astro static site with project cards, a timeline, Markdown blog, light/dark themes and a generative background; deployed with GitHub Actions to GitHub Pages on a custom domain

HONORS & ACHIEVEMENTS

- GENIUS Olympiad 2026 Finalist & Stage 2 Stockholm Junior Water Prize - Artificial Intelligence category, for AcousticLeakNet
- Competition wins: 1st - DPS Saket International Hackathon; 1st - DPS RK Puram Hardware Competition; 3rd - DPS RK Puram Game Development; 3rd of 50 DPS branches - Inter DPS Environment Fest
- Won 30+ regional and national events with eSpice (IT Club, member for 4 years); participated in 50+ interschool competitions (past 3 years)
- President of eSpice (2026-27), Joint Secretary of eSpice 2025-26; mentoring students in AI/ML, Web Dev, Game Dev and Hardware. Active member, science and technology clubs, DPS Noida
- Promoted to the Post Graduate Level in Gurugram Police Cyber Security Summer Internship (GPCSSI)

SKILLS

Programming: Python, C/C++, JavaScript, HTML/CSS, SQL, Bash, PowerShell
AI / ML: Deep learning, PyTorch, TensorFlow, data analysis (pandas), signal processing
Tools: Git/GitHub, GitHub Actions, LaTeX, QEMU, YARA

INTERESTS

Software engineering, applied AI, network and systems security, and the hardware and servers that everything runs on.